

**AI-DRIVEN ELIGIBILITY PREDICTION AND
HEALTHCARE RECOMMENDATION SYSTEM FOR
DIGITAL WELFARE MANAGEMENT IN SRI LANKA**

25-26J-320

Project Proposal Report

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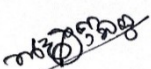
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Declaration

I hereby declare that this proposal is my own work and does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where due reference is made in the text.

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Abstract

In Sri Lanka, millions of people rely on social welfare programs like Samurdhi, disability allowances, and pensions to meet their basic needs. However, the current welfare system faces many challenges, such as slow manual processes, fragmented data, and difficulties ensuring the right people receive help. These issues often cause delays and prevent vulnerable citizens from getting timely support. This research proposes a new AI-powered digital platform to transform how welfare services are managed and delivered across the country. By using data about people's economic, social, and health situations, the system will smartly predict who qualifies for aid and suggests appropriate healthcare services tailored to individual needs. It will also include helpful features like a multilingual chatbot and real-time tracking to make the experience easier for users. Inspired by successful e-governance models from other countries, this project aims to bring fairness, speed, and transparency to welfare distribution in Sri Lanka. The goal is to build a trusted, user-friendly platform that not only improves access for citizens but also equips policymakers with better tools to make data-driven decisions. Ultimately, this research hopes to make social welfare more inclusive, efficient, and responsive to the real needs of the people.

Keywords: Artificial Intelligence, Welfare Services, Eligibility Prediction, Healthcare Recommendation, Sri Lanka

Table of contents

Table of Contents

Declaration	iii
Abstract	iv
Table of Contents.....	v
Table of Figures	vi
List of Tables.....	vi
List of Abbreviations.....	vi
List of Appendices.....	vi
1. Introduction	1
1.1. Background & Literature Review	3
1.2. Research Gap.....	6
1.3. Research Problem.....	8
2. Main and Sub Objectives	9
2.1 Main Objectives	9
2.2 Sub Objectives.....	9
3. Methodologies	10
3.1 Overall System	10
3.2 Individual System Diagram.....	11
4. Gantt Chart and Work Breakdown Structure.....	12
5. Project Requirements	13
5.1 Functional Requirements.....	13
5.2 Non-Functional Requirements.....	14
6. Commercialization	15
6.1 Budget.....	15
6.2 Commercialization Methods	16
7. References	17
8. Appendices.....	18
Appendix A: Similarity percentage result in Turnitin.....	18

Table of Figures

Figure 1.0: Impact of Digital Welfare Platforms on Service Reach and Satisfaction in South Asia.....	2
Figure 2.0: AI-powered Digital Welfare Platform Architecture.....	10
Figure 3.0: Eligibility Prediction and Healthcare Recommendation System Diagram	11
Figure 4.0: Gantt Chart	12
Figure 4.1: Work Breakdown Structure.....	12

List of Tables

Table 1.0: Comparison Between Proposed Platform and Existing Systems.....	7
Table 2.0: Budget Breakdown	15

List of Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
DBT	Direct Benefit Transfer
D2C	Direct-to-Consumer
FAQ	Frequently Asked Questions
GIS	Geographic Information System
ICTA	Information and Communication Technology Agency (Sri Lanka)
ML	Machine Learning
NLP	Natural Language Processing
SQL	Structured Query Language
SSL	Secure Sockets Layer
VPS	Virtual Private Server

List of Appendices

Appendix A: Similarity percentage result in Turnitin	18
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1. INTRODUCTION

This research project aims to develop an AI-powered digital platform to improve the efficiency and fairness of social welfare services management in Sri Lanka. Currently, Sri Lanka offers a variety of welfare programs such as Samurdhi, disability allowances, elderly pensions, and nutritional aid. However, the existing system faces several critical challenges, including manual application handling, fragmented data across agencies, lack of real-time tracking, and delays in approvals. These inefficiencies often result in deserving citizens facing barriers to access benefits, while some ineligible individuals receive aid.

The COVID-19 pandemic further highlighted the urgent need for digital transformation in public services worldwide. Many countries successfully implemented e-governance solutions to improve transparency, accessibility, and operational efficiency of welfare programs. Sri Lanka, however, has yet to adopt a comprehensive, AI-enabled digital welfare platform that integrates smart eligibility checks, real-time monitoring, and multilingual citizen support.

The proposed solution is designed to enhance, not replace, existing welfare processes by providing a centralized, user-friendly system that benefits citizens, Divisional Secretariat officers, and policymakers alike. The platform is composed of three main modules:

- **Citizen Portal:** Enables citizens to check eligibility through an AI-powered prediction engine, access a multilingual chatbot for assistance, track application status in real-time, upload necessary documents, and schedule appointments.
- **Officer Dashboard:** Allows government officers to centrally manage and verify applications, reducing manual processing delays and improving accuracy.
- **Policy Admin Panel:** Equips policymakers with AI-powered dashboards featuring real-time data visualization, fraud detection, risk mapping, and automated insights for informed decision-making.

A 2020 study by the World Bank, hosted by the Abdul Latif Jameel Poverty Action Lab (J-PAL), demonstrated that digital welfare platforms in South Asia increased service reach and beneficiary satisfaction by over 40% compared to traditional methods (Figure 1). This research leverages similar approaches, aiming to significantly improve welfare delivery outcomes in Sri Lanka

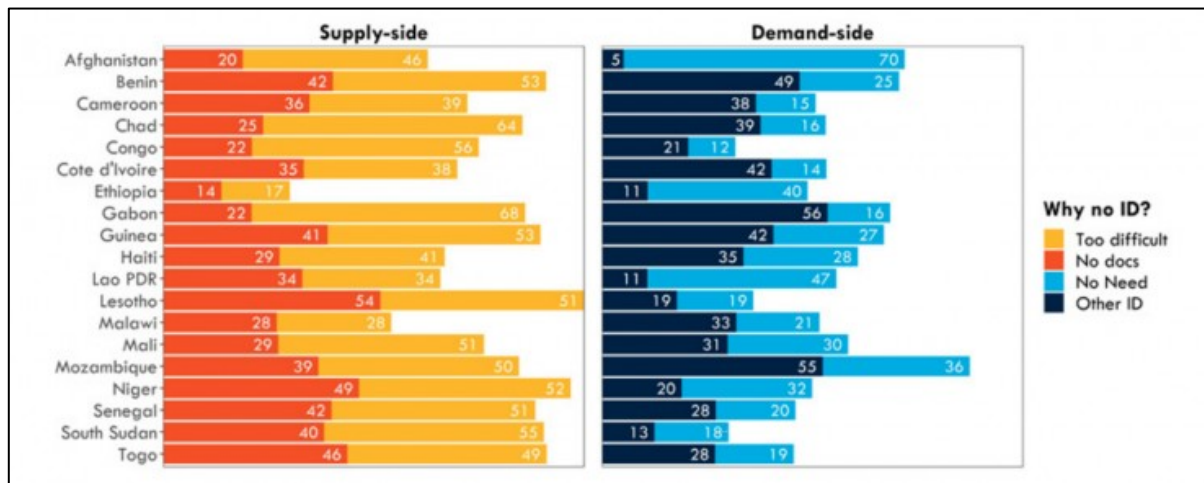


Figure 1.0: Impact of Digital Welfare Platforms on Service Reach and Satisfaction in South Asia

My contribution focuses on developing the AI-powered Eligibility Prediction Engine and Healthcare Recommendation System. This component will analyze socio-economic and health data to accurately assess citizens' eligibility for various welfare programs and recommend appropriate healthcare services based on individual needs. By integrating machine learning classification models and ranking algorithms, this system will help ensure resources are distributed fairly and efficiently. The goal is to empower both citizens and officers with intelligent tools that simplify welfare access and administration, ultimately enhancing social equity and well-being.

1.1 Background & Literature Review

Social welfare programs are essential mechanisms designed to support the vulnerable and disadvantaged segments of society. In Sri Lanka, programs like Samurdhi, disability allowances, elderly pensions, and nutritional aid have played a critical role in poverty alleviation and social protection. Despite these efforts, the current welfare management system suffers from several longstanding challenges that limit its effectiveness. Applications are often handled manually, leading to delays and errors. Data remains fragmented across different government departments, making real-time tracking and verification difficult. This fragmented approach results in some eligible beneficiaries facing significant barriers in accessing benefits, while others who are not qualified still manage to receive aid due to loopholes and inefficiencies [1][4].

Digital transformation offers a promising pathway to overcoming these barriers. Globally, governments have begun adopting e-governance models that leverage technology to improve welfare service delivery. For example, India's Aadhaar system, a biometric ID platform, has revolutionized welfare disbursement by enabling quick identity verification, reducing duplication, and increasing transparency [2]. Bangladesh has similarly integrated digital technologies to improve welfare accessibility and reduce corruption in its programs [3]. These countries have seen notable improvements in both the reach and satisfaction of their welfare services.

However, Sri Lanka's welfare system remains largely dependent on traditional, manual methods and lacks a unified, intelligent platform that can incorporate AI and real-time data analysis [4]. The absence of such a platform leads to inefficiencies including administrative delays, duplication of applications, and difficulties in monitoring and evaluating welfare outcomes.

The integration of Artificial Intelligence (AI) and machine learning techniques has the potential to address many of these challenges. Machine learning classification models such as decision trees, random forests, and logistic regression can analyze complex socio-economic and demographic datasets to predict program eligibility accurately. These predictive models reduce human error and prevent resource leakage, ensuring aid reaches the right beneficiaries [5]. Multilingual AI-powered chatbots further increase accessibility by providing instant, 24/7 assistance in various local languages, thereby bridging communication gaps especially in rural and underserved communities [6]. Geographic Information Systems (GIS) combined with AI can help policymakers visualize high-risk areas, detect fraud, and plan resource allocation more efficiently [7].

Advantages of AI-powered digital welfare platforms include:

- **Enhanced Accuracy in Eligibility Assessment:** AI models analyze large volumes of data to identify eligible beneficiaries with high precision, reducing both under-coverage and leakage [1][5].
- **Improved Accessibility:** Multilingual chatbots and mobile-friendly portals ensure citizens from diverse linguistic backgrounds and regions can easily access welfare services without physical visits [6].
- **Transparency and Accountability:** Real-time tracking and digital records increase transparency, minimizing corruption and building public trust in welfare programs [8].
- **Operational Efficiency:** Automated workflows reduce manual processing time, enabling faster approvals and disbursement of benefits [4].
- **Policy Support:** AI-driven dashboards provide policymakers with actionable insights, enabling evidence-based decision-making and better resource planning [7].

Despite these benefits, the shift toward digital welfare platforms presents several challenges and limitations:

- **Digital Divide:** Limited internet penetration and low digital literacy, particularly in rural areas, may exclude certain populations from fully benefiting [9].
- **Privacy and Data Security:** Handling sensitive personal and financial data requires stringent safeguards to prevent breaches and misuse [10].
- **Resistance to Change:** Government officials and beneficiaries accustomed to manual processes may resist transitioning to digital platforms, requiring comprehensive training and awareness campaigns [11].
- **Infrastructure Limitations:** Inadequate hardware, network connectivity issues, and lack of integration across departments hinder effective implementation [9].
- **Language and Cultural Barriers:** While multilingual support helps, accurately capturing local dialects and cultural nuances remains challenging for AI chatbots [6].

Existing digital welfare platforms globally provide valuable lessons and highlight gaps that Sri Lanka's system must address. Popular platforms like India's Direct Benefit Transfer (DBT) have shown how centralized databases and biometric verification can reduce fraud. However, these systems often lack personalized support features such as AI-powered recommendations or interactive assistance for beneficiaries struggling with applications [2][3]. Moreover, many platforms do not integrate healthcare recommendations, which are critical for vulnerable groups with health needs. Multilingual AI chatbots are emerging but are not yet widespread or sufficiently localized [6].

This research addresses these gaps by proposing a comprehensive digital welfare platform tailored to Sri Lanka's unique social, economic, and linguistic context. Key components include:

- **AI-powered Eligibility Prediction Engine:** Uses machine learning to assess eligibility based on socio-economic data, improving accuracy and reducing manual workload [5].
- **Healthcare Recommendation System:** Suggests appropriate healthcare services based on individual health profiles, bridging welfare and healthcare support [1].
- **Multilingual Chatbot:** Provides accessible assistance to beneficiaries in their native languages, simplifying communication and reducing barriers [6].
- **Real-time Tracking and Policy Dashboards:** Facilitates transparency and informed decision-making through dynamic data visualization and fraud detection [7][8].

In summary, integrating AI into welfare management presents a unique opportunity for Sri Lanka to enhance social equity by ensuring aid is efficiently and fairly distributed. While challenges remain, a carefully designed digital platform can transform welfare delivery and improve lives at scale.

1.2 Research Gap

A research gap refers to an area within a field that has not been thoroughly explored or addressed in previous studies. In the context of social welfare services in Sri Lanka, several critical gaps remain unaddressed despite ongoing efforts to improve program delivery. These gaps create challenges for beneficiaries, government officers, and policymakers alike.

One of the most significant gaps is the absence of intelligent and automated eligibility prediction in existing welfare systems. Currently, eligibility assessments often rely on manual review or basic rule-based checks. This approach is prone to errors such as wrongly excluding eligible citizens or providing benefits to ineligible individuals. The lack of AI-based predictive models means that resources may not be optimally allocated, reducing the overall fairness and efficiency of welfare programs. Incorporating machine learning techniques could enhance the accuracy of eligibility decisions by analyzing complex socio-economic and demographic data, which traditional methods cannot fully leverage.

Another major gap is the lack of real-time, multilingual support for beneficiaries. Many citizens, particularly those in rural areas or with limited digital literacy, find it difficult to access welfare information and complete application processes. Existing platforms rarely provide interactive assistance in local languages such as Sinhala and Tamil. This linguistic barrier leads to confusion, application errors, and ultimately limits the reach of welfare services. AI-powered chatbots capable of understanding and communicating in multiple languages could significantly improve accessibility and user experience.

Additionally, policymakers face a gap in accessing timely and actionable data. Current systems lack integrated dashboards that provide real-time visualizations of welfare program usage, fraud detection, and risk mapping. Without such tools, it is difficult for decision-makers to identify problem areas, prevent misuse of resources, and plan effectively. AI and Geographic Information Systems can support dynamic monitoring, enabling data-driven policy formulation and better allocation of funds.

Furthermore, there is a lack of integration between welfare services and healthcare recommendations. Vulnerable groups often require not only financial aid but also health support. Existing welfare platforms do not offer personalized healthcare guidance based on individual health profiles. This gap reduces the holistic effectiveness of welfare programs and limits support for beneficiaries with medical needs.

The table below summarizes the key gaps by comparing the proposed AI-powered welfare platform with current welfare systems in Sri Lanka:

Table 1.0: Comparison of Key Features Between Proposed AI-Powered Welfare Platform and Existing Sri Lanka Welfare Systems

Feature	Proposed Platform	Current Sri Lanka Welfare Systems (Manual System)
Personalized Healthcare Recommendation	Yes	No
Multilingual Chatbot Support	Yes	No
AI-based Eligibility Prediction	Yes	No
Real-time Application Tracking	Yes	Partial
AI-powered Fraud Detection and Risk Mapping	Yes	No
Policy Dashboard with Predictive Analytics	Yes	No

By addressing these research gaps, the proposed platform aims to enhance fairness, accessibility, and operational efficiency in Sri Lanka's welfare services. It combines cutting-edge AI technologies with user-centered design to meet the unique social, linguistic, and administrative challenges of the country. This research will contribute to bridging the divide between traditional welfare practices and the digital transformation needed for future sustainability and equity.

1.3 Research Problem

The research problem identifies key challenges in the current social welfare system in Sri Lanka that this study aims to address. These problems revolve around inefficiencies, lack of automation, and limited accessibility, which prevent the welfare system from serving its beneficiaries effectively. The main research problems identified are:

- **How to accurately predict eligibility for welfare programs using AI:** The current manual or rule-based eligibility assessments are prone to errors, leading to eligible citizens being excluded and ineligible ones receiving benefits. The challenge lies in developing AI models that can analyze complex socio-economic and demographic data to make fair and accurate eligibility predictions. This involves selecting appropriate machine learning techniques and ensuring that the system adapts to diverse citizen profiles.
- **How to provide real-time, multilingual support to beneficiaries:** Many welfare applicants struggle to understand application requirements or track their status due to language barriers and limited digital literacy. Designing an AI-powered chatbot that can communicate fluently in Sinhala, Tamil, and English is essential to bridge this communication gap and make welfare services more accessible and user-friendly for all citizens.
- **How to enable timely decision-making through dynamic data visualization and fraud detection:** Policymakers currently lack tools to monitor welfare program usage, identify fraud, or visualize risk areas in real-time. Developing AI-driven dashboards with geographic information and predictive analytics is necessary to provide actionable insights that improve transparency, reduce resource misuse, and optimize resource allocation.
- **How to integrate healthcare recommendations into welfare services:** Vulnerable beneficiaries often require healthcare support alongside financial aid, but existing systems do not offer personalized healthcare guidance. The challenge is to build a recommendation system that considers individual health profiles and connects citizens with appropriate medical services, enhancing the overall welfare impact.

By addressing these research problems, the proposed platform aims to create a more efficient, inclusive, and intelligent welfare management system that better serves the people of Sri Lanka.

2. MAIN AND SUB OBJECTIVES

2.1 Main Objectives

The main objective of this research is to develop an AI-powered digital welfare platform that enhances the fairness, accessibility, and efficiency of social welfare service delivery in Sri Lanka. This platform aims to support citizens, government officers, and policymakers by providing intelligent eligibility prediction, personalized healthcare recommendations, and real-time service monitoring. By leveraging advanced AI techniques, the research intends to streamline welfare processes, reduce resource misallocation, and improve overall beneficiary satisfaction.

To achieve this main goal, the research focuses on the following sub objectives:

- Develop an AI-based eligibility prediction engine to accurately assess citizen eligibility for various welfare programs.
- Create a healthcare recommendation system that suggests suitable health services based on individual health profiles.
- Implement a multilingual chatbot to assist beneficiaries in accessing welfare information and completing applications easily.
- Design real-time tracking and policy dashboards for officers and policymakers to monitor welfare distribution and detect fraud efficiently.

2.2 Sub Objectives

My specific responsibility within this project is to design and implement the AI-powered Eligibility Prediction Engine and Healthcare Recommendation System. To accomplish this, I have broken down the work into the following sub objectives:

- **Collect and preprocess socio-economic and health data:** Gather relevant datasets on citizen demographics, income, health status, and program participation to train accurate predictive models.
- **Develop machine learning classification models:** Build and train models such as decision trees and logistic regression to evaluate eligibility based on complex data patterns.
- **Design the healthcare recommendation algorithm:** Create a system that analyzes individual health profiles and suggests appropriate hospitals, specialists, and medical services, improving beneficiary health support.
- **Integrate AI components with the platform:** Ensure seamless integration of eligibility prediction and healthcare recommendations into the citizen portal for easy access and real-time feedback.

3. METHODOLOGIES

3.1 Overall System

The proposed AI-powered digital welfare platform is designed to improve the efficiency, fairness, and accessibility of social welfare services in Sri Lanka. The system is composed of four main components that work together to streamline welfare delivery and provide intelligent support to citizens, officers, and policymakers. These components are:

- **Eligibility Prediction Engine:** Uses machine learning models to analyze socio-economic and demographic data to accurately predict citizens' eligibility for various welfare programs, reducing manual errors and increasing fairness.
- **Healthcare Recommendation System:** Provides personalized healthcare suggestions by analyzing individual health profiles, connecting beneficiaries with appropriate medical services and specialists.
- **Multilingual Chatbot Support:** Offers interactive, 24/7 assistance in Sinhala, Tamil, and English to guide users through eligibility checks, application processes, and FAQs, improving accessibility especially for rural and underserved populations.
- **Policy Dashboard and Fraud Detection:** Delivers real-time data visualizations and AI-powered fraud detection tools for government officers and policymakers to monitor welfare distribution, identify anomalies, and make informed decisions.

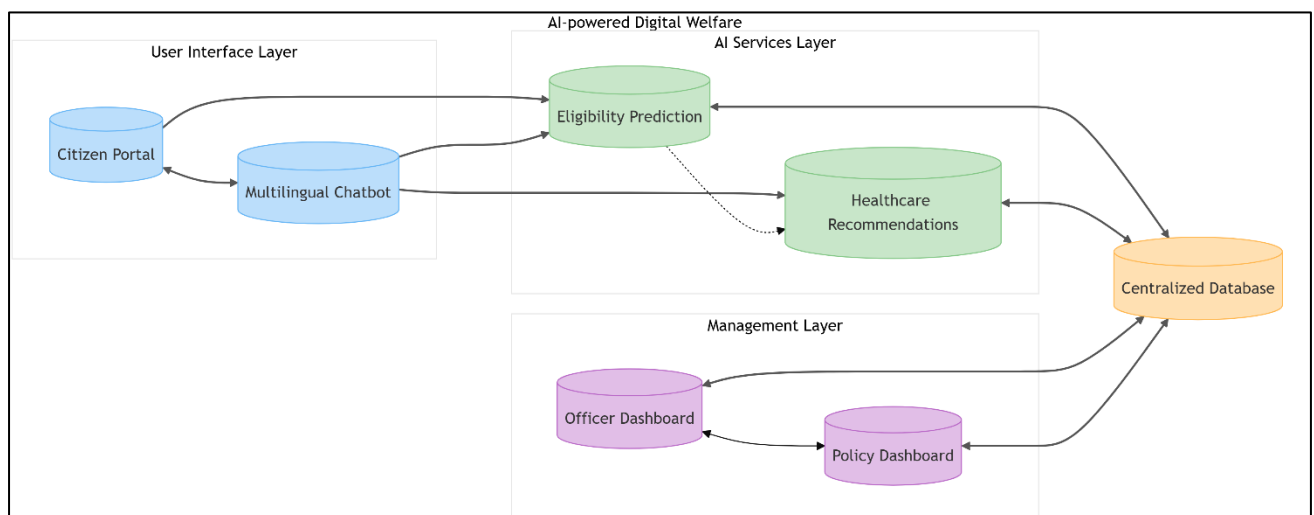


Figure 2.0: AI-powered Digital Welfare Platform Architecture

3.2 Individual System Diagram

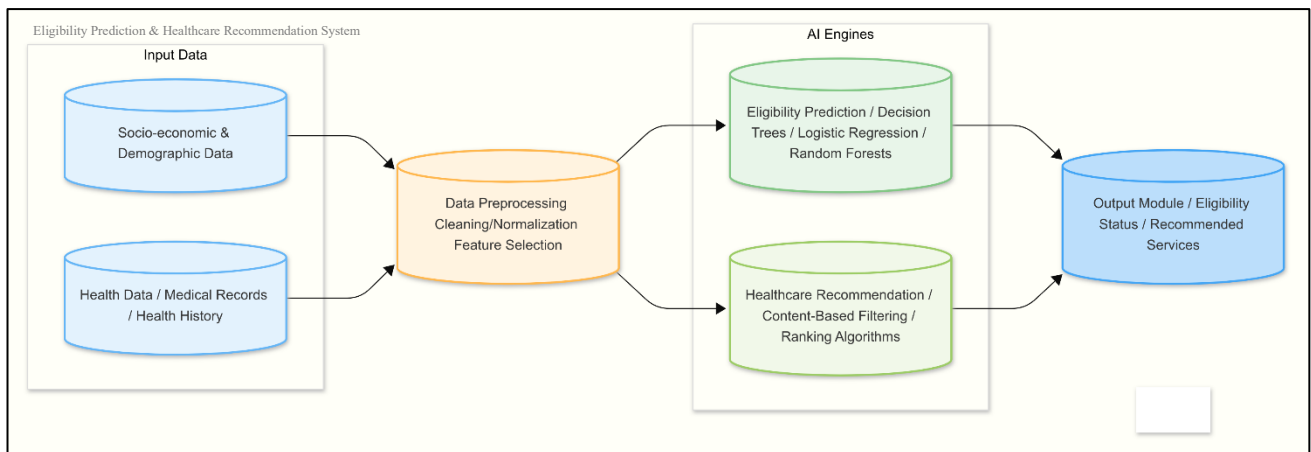


Figure 3.0: Eligibility Prediction and Healthcare Recommendation System Diagram

This component, which I am responsible for developing, integrates eligibility prediction and healthcare recommendations to empower welfare officers and citizens with timely, accurate information.

The system is divided into four subcomponents:

Subcomponent to assign patients with activities to do during the post therapy period: By consulting therapists involved in online therapy sessions, we will be identifying activities that are proven to reduce stress of patients than make it worse depending on certain scenarios. Then these activities will be assigned to patients to carry out during the post therapy period of the patient.

- **Data Collection and Preprocessing:** Gathers and prepares socio-economic and health data from multiple sources, ensuring quality and consistency for machine learning models.
- **Eligibility Prediction Module:** Applies classification algorithms such as decision trees and logistic regression to assess eligibility based on complex data patterns.
- **Healthcare Recommendation Module:** Uses ranking algorithms to suggest suitable healthcare services tailored to the beneficiary's health profile and needs.
- **Integration and User Interface:** Connects the AI models to the citizen portal and officer dashboards, providing intuitive access and real-time feedback to end-users

4. GANTT CHART AND WORK BREAKDOWN STRUCTURE

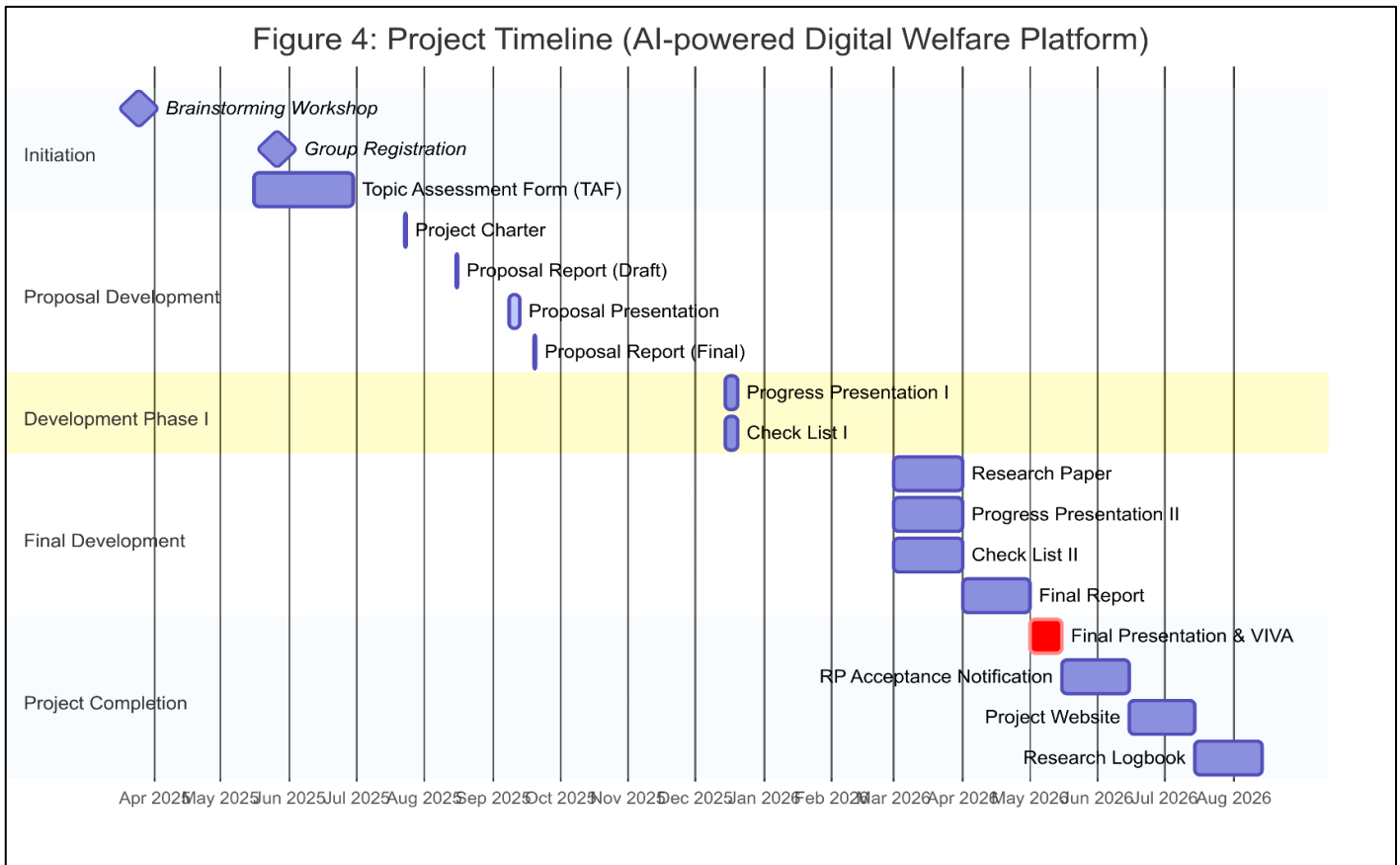


Figure 4.0: Gantt Chart

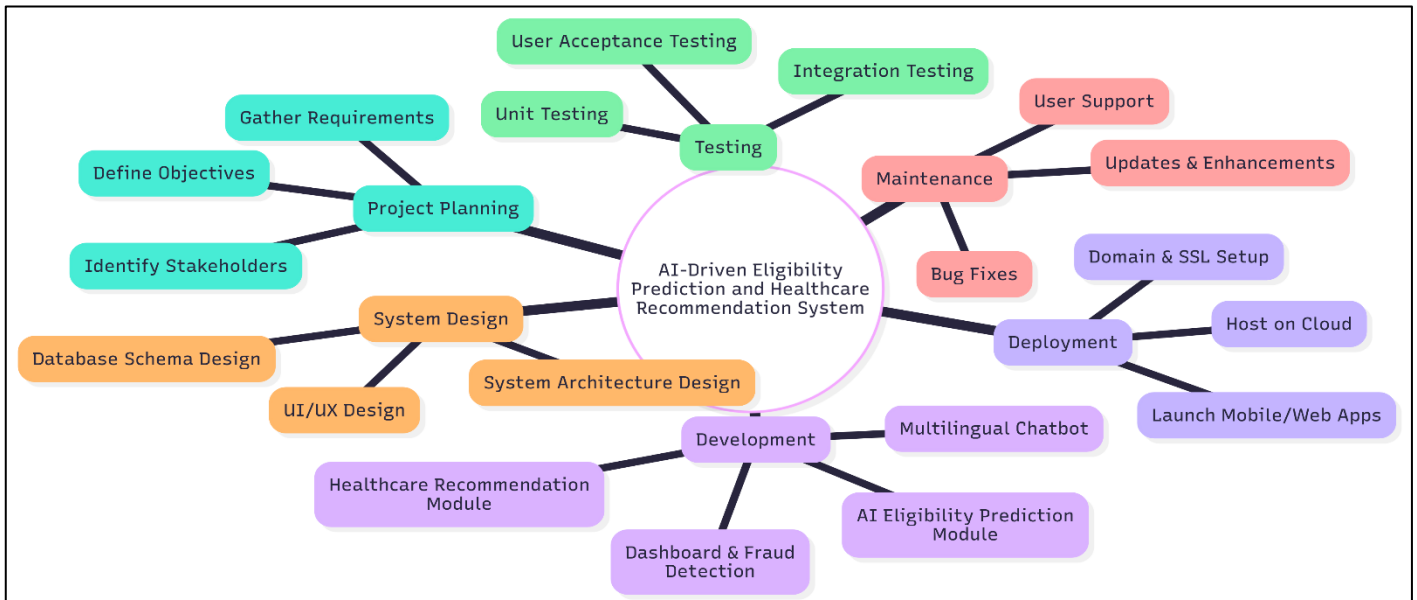


Figure 4.1: Work Breakdown Structure

5. PROJECT REQUIREMENTS

5.1 Functional Requirements

The functional requirements define the specific operations, responsibilities, and capabilities that the proposed AI-powered Digital Welfare Platform must perform. These requirements outline the expected inputs, outputs, and system behaviors under various conditions to ensure that the platform effectively supports citizens, welfare officers, and administrators.

The functional requirements for the system include:

- **Citizen Application Submission:** The system should enable citizens to submit welfare program applications online, including personal details, eligibility-related documents, and healthcare information.
- **Automated Eligibility Prediction:** Using AI algorithms, the system should analyze the submitted application data and predict the applicant's eligibility status for welfare programs.
- **Healthcare Recommendation:** Based on the applicant's medical and demographic data, the system should generate personalized healthcare recommendations and connect them to available public health services.
- **Application Tracking:** The system should allow applicants to track the real-time status of their submissions, from initial review to approval or rejection.
- **Officer Dashboard:** Welfare officers should be able to review AI-generated eligibility predictions, make necessary verifications, and update application statuses.
- **Notification & Alerts:** The system should send automated notifications to applicants regarding status updates, required document submissions, or recommended healthcare services.

5.2 Non-Functional Requirements

Non-functional requirements specify the quality attributes and operational conditions under which the system must operate. These include performance, scalability, security, usability, and reliability aspects.

- **Security:** Ensure that all applicant data, including personal and medical information, is encrypted and protected from unauthorized access.
- **Reliability:** The system should be highly available and accessible to citizens and welfare officers without downtime during working hours.
- **Scalability:** The platform should be able to handle a large number of concurrent users during peak welfare application periods.
- **Efficiency:** AI algorithms should process and return eligibility predictions and recommendations within a few seconds.
- **Usability:** The interface should be intuitive and easy to navigate for users with varying technical skills.

6. COMMERCIALIZATION

6.1 Budget

The following budget table provides an estimated cost breakdown for launching and maintaining the proposed Eligibility Prediction and Healthcare Recommendation System. The costs are approximate and may vary depending on vendor pricing, system scale, and chosen service providers.

Table 2.0: Budget Breakdown

Description	Cost	Occurring
Server Hosting (Cloud VPS)	LKR 10,000	Monthly
App Store Hosting (iOS Developer Program)	LKR 7,754	Annual
Play Store Hosting (Google Play Developer)	LKR 30,700	One-Time
Cloud Database Storage (Amazon S3 / Firebase)	0.023USD per GB	Monthly
Domain Registration & SSL Certificate	LKR 5,000	Annual
Third-Party API Costs (Health Data, AI Models)	LKR 15,000	Monthly
Marketing & Promotion	LKR 50,000	Initial Launch
Maintenance & Technical Support	LKR 25,000	Monthly

6.2 Commercialization Methods

The system will be introduced as a scalable healthcare technology solution that can be adopted by both public and private healthcare providers. Our commercialization strategy focuses on B2B partnerships and direct-to-consumer (D2C) models:

1. B2B Partnerships:

- Collaborate with hospitals, clinics, and telemedicine platforms to integrate the system into their patient management processes.
- Offer subscription-based licensing for institutions with tiered pricing based on patient volume and feature access.

2. Direct-to-Consumer (D2C):

- Provide a mobile application for patients and a web dashboard for healthcare professionals.
- Use a freemium model:
 - Free Tier: Basic eligibility checks, general healthcare recommendations, and limited access to health data storage.
 - Premium Tier: Advanced AI predictions, personalized treatment recommendations, priority appointment scheduling, and detailed analytics reports.
- Monthly subscription plans for individuals and healthcare professionals.

3. Market Expansion:

- Launch initially in Sri Lanka targeting urban and semi-urban regions where digital healthcare adoption is growing.
- Expand regionally into South Asian countries with similar healthcare challenges.

4. Revenue Streams:

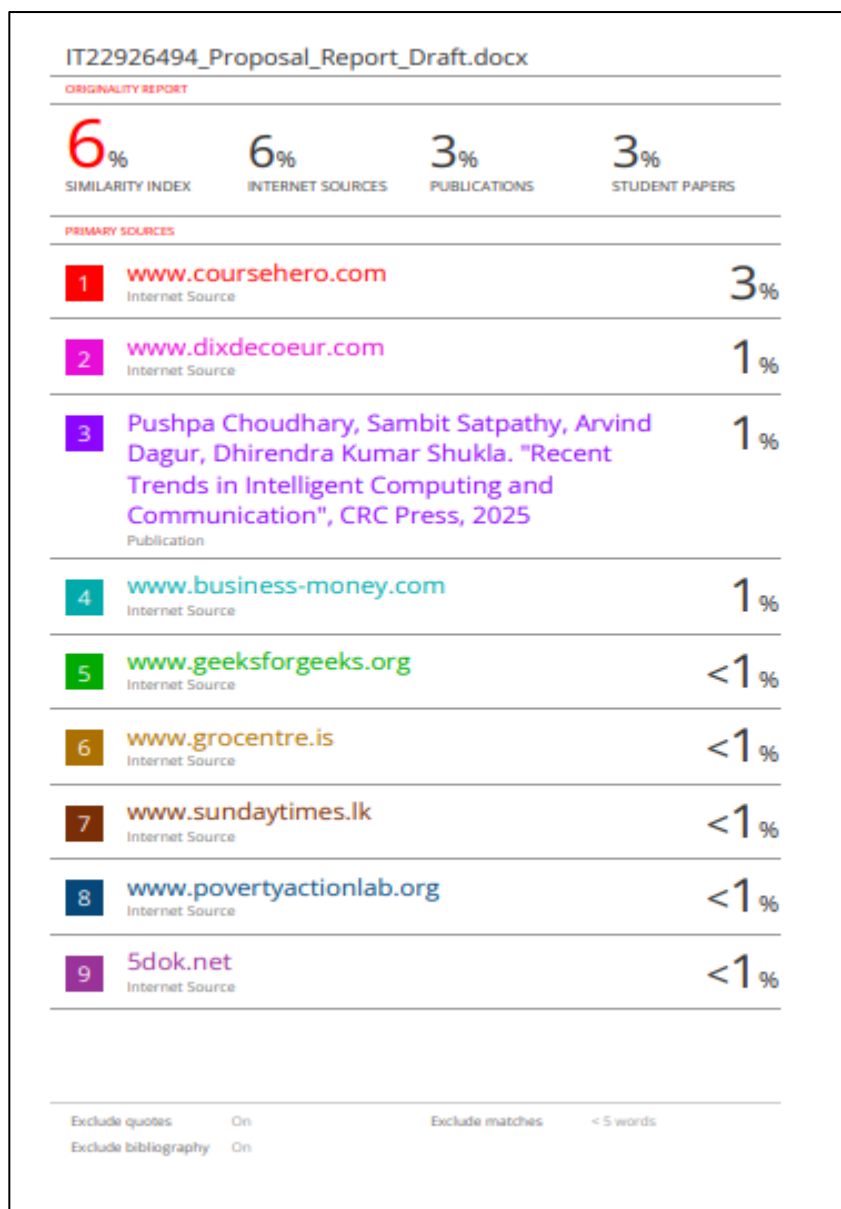
- Subscription fees (individual & institutional).
- Premium analytics and reporting services.
- Licensing of AI prediction models to third-party platforms.
- Data insights for health research (in compliance with privacy laws).

By combining a freemium approach with institutional partnerships, the system can achieve early adoption, recurring revenue, and sustainable growth while contributing to improved healthcare accessibility and efficiency.

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8. APPENDICES



Appendix A: Similarity percentage result in Turnitin

The proposal document was tested using the Turnitin plagiarism detection system to ensure academic integrity and originality. As illustrated in Appendix A, the similarity index was identified as 6%, which falls well below the accepted threshold of 15%. This confirms that the content is primarily original and properly cited, meeting the university's submission standards.